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### Method and device for providing a licence plate, preferably a motor vehicle licence plate, with a data carrier

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#### Abstract

A device for the automatic fitting of vehicle registration plates with data carriers. A plurality of work stations interlinked by at least one conveyor are provided for this purpose. The work stations are equipped with handling devices and associated imaging devices. As a result, a respective data carrier can be automatically gripped and placed in its precise position in the opening and prefixed there. The data carriers are then each cast automatically in the opening for the permanent fixation of the data carrier in the opening of the registration plate.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 16/653,376 having a filing date of 27 Mar. 2020, which is the US National Phase of and claims the benefit of and priority on International Application No. PCT/EP2018/000451 having a filing date of 26 Sep. 2018, which claims priority on and the benefit of German Patent Application No. 10 2017 009 108.0 having a filing date of 29 Sep. 2017 and German Patent Application No. 10 2018 004 849.8 having a filing date of 20 Jun. 2018.

## BACKGROUND OF THE INVENTION

### Technical Field

(1) The invention relates to a method for providing a registration plate, preferably a motor vehicle license plate, with at least one data carrier that can be read contactlessly, wherein the data carrier is fixed in a receptacle, provided for this purpose, of a plate body or of a material web for the formation of plate bodies, or wherein the or the respective data carrier is positioned in a receptacle, provided for this purpose, of a plate body or of a material web for the formation of plate bodies of the registration plate. Further, the invention relates to a device for providing a registration plate, preferably a motor vehicle license plate, with at least one data carrier that can be read contactlessly, with at least one work station and with at least one plate body having at least one receptacle for a data carrier or with a material web for the formation of plate bodies of the registration plate.

### Prior Art

(2) Registration plates, primarily and specifically motor vehicle license plates, are increasingly provided with data carriers that can be read contactlessly, and preferably also written contactlessly. (3) The data carriers must be positioned precisely on or in a receptacle provided for this purpose in the plate body of the registration plate or in successive receptacles in a material web from which the plate body is punched in order for the data carriers to operate smoothly and in particular that they can be read. The ever-increasing need for registration plates provided with data carriers necessitates an automation of the process of applying the data carriers to the plate body or to material webs. No corresponding methods or devices are hitherto known.

## BRIEF SUMMARY OF THE INVENTION

(4) It thus follows that the invention is based on the object of providing precise methods and devices which are fully automatic at least to the greatest possible extent, by means of which registration plates, preferably motor vehicle license plates, or blanks and/or material webs for the formation of same, can be provided with data carriers which can at least be read contactlessly. (5) One method for achieving this object is a method for providing a registration plate, preferably a motor vehicle license plate, with at least one data carrier which can be read contactlessly, wherein the data carrier is fixed in a receptacle, provided for this purpose, of a plate body or of a material

web for the formation of plate bodies, characterized in that the data carrier is pre-fixed in the predetermined position relative to the plate body or to the material web and/or to the receptacle in the plate body or material web and is then finally fixed in this relative position. Accordingly, it is provided that the or the respective data carrier is initially pre-fixed in its intended position relative to the plate body or in a material web for the formation of the plate body, in particular in a receptacle provided for this purpose, and subsequently finally fixed in this relative position. As a result, the respective data carrier can be automatically attached exactly in or on the respective receptacle of the plate body or material web.

(6) The final fixing of the data carrier on or in the respective receptacle is preferably executed by casting or at least recasting in the or each receptacle. This creates a permanently reliable attachment of the data carrier in the plate body or in the material web for forming registration plate bodies. A data carrier attached in such a manner to the registration plate can no longer be separated from the registration plate without destroying it. This is particularly important in the case of motor vehicle license plates.

(7) The pre-fixing of the data carrier on or in the respective receptacle of the plate body or material web is preferably made by means of a single or a small number of adhesion sites having a small surface area. The at least one adhesion site is applied in the region of the receptacle for the data carrier. The at least one small-area adhesion site is preferably applied or inserted into the receptacle before the data carrier is pre-fixed on the receptacle. Particular preferred for this purpose is a central and/or centred region of the receptacle. The pre-fixing process stabilises the data carrier that has been placed in the receptacle prior to and until the final fixation has been completed. Here the at least one adhesion site ensures that the position of the data carrier on the receptacle or in the receptacle does not change, or does not appreciably change, until the data carrier is fixed in the receptacle and during this fixation. One advantageous possible further development of the method offers additional support for the positioning of the adhesion site at the intended receptacle for the data carrier by providing an imaging device. Instead of the imaging device, a measuring device can also be provided, in particular when a material web is to be provided with data carriers. This option also applies in the following when reference is made only to imaging devices.

(8) The position values from the receptacle generated by the imaging device or the like preferably control the application of the at least one adhesion site. As a result, the respective adhesion site can already be applied on or in the receptacle precisely at the intended position, specifically in a preferred automatic manner by means of a handling device, for example, or simply by a dispenser device for a substance for forming the adhesion site.

(9) In a particularly advantageous embodiment of the method, where a plate body or material web is provided on one side with a film at least in the region of the respective receptacle, and one or each receptacle for a data carrier is formed as an opening, provision is made to apply the at least one adhesion site in the region of the respective opening on the film, or to employ the self-adhesive film for attaching the respective data carrier as the adhesion site. The film thus serves practically as a carrier for the at least one adhesion site in every continuous receptacle, which is open on all sides, in the plate body or in the material web, or that the film itself forms the adhesion site. In case the receptacle is formed as a depression, the plate body or material web does not necessarily have to be provided or covered with a film.

(10) A further method for achieving the object stated at the outset, whereas this may also be a preferred further development of the previously described method, is a method for providing a registration plate, preferably a motor vehicle license plate, with at least one data carrier which can be read contactlessly, wherein the or the respective data carrier is positioned in a receptacle, provided for this purpose, of a plate body or of a material web for the formation of plate bodies of the registration plate, characterized in that the at least one data carrier is gripped by a handling device in a targeted manner, the position of the or each of the data carriers held by the handling device is determined by a measuring or imaging device and the plate body or the material web is

then fitted with at least one data carrier on the basis of the obtained position of the data carrier on the handling device. Accordingly, in each case a single or, if necessary, a plurality of data carriers are gripped in a targeted manner by a handling device, with the position of the respective data carrier held by the handling device being determined by an imaging device so that, on the basis of the thereby received data, in particular position data, of the data carrier on the handling device, the respective data carrier is inserted into the respective receptacle of the plate body or material web which is held ready at a specific position relative to the handling device. Since the imaging device has determined the position of data carrier after it has been gripped by the handling device, the exact position of the data carrier on the handling device is known from the point in time when the imaging device has acquired its data. Based on this known position, the handling device can then deposit the data carrier precisely in or on the receptacle. The position of the receptacle is known from the specific position relative to the handling device for holding ready the plate body with the receptacle or the continuous material web with successive receptacles on defined positions of same. The method makes it possible to achieve a highly precise positioning of the data carrier on or in the respective receptacle.

(11) Provision can be made for the data carrier to be pre-fixed in the position in which it is put on or inserted in the receptacle by at least one adhesion point previously applied in or on the receptacle. As a result, the data carrier inserted by the handling device into the receptacle is simultaneously pre-fixed in or on the receptacle. The data carrier therefore maintains the position in which it was brought to the receptacle. The data carrier cannot slip out of place in or on the receptacle during further transport, if necessary, to a subsequent work station and/or during the final step of complete fixation in or on the receptacle.

(12) According to a further possible development of the method, provision is made that an imaging device supports, and preferably controls, the handling device in the preferably targeted gripping in each case of preferably a single data carrier, but if necessary a number of data carriers simultaneously, from a plurality of data carriers which preferably lie ready in a flat position on a level surface. As a result, the handling device can reliably travel up to the position of a particularly favourably situated data carrier in the store of allocated data carriers and securely grip the or each selected data carrier whose position has been determined. Thus, in each case the handling device, supported by the imaging device, “picks out” one or each data carrier from a store comprising a plurality of data carriers.

(13) One further possible development of the method provides for verifying the or each data carrier held by the handling device after a data carrier has been gripped by the handling device. This ensures that no receptacle is occupied by an inoperative data carrier.

(14) One further development of the method, which may also represent an independent solution of the object on which the invention is based, is characterized in that after the at least one data carrier has been gripped by the handling device at least individual data are read from the data carrier and preferably stored in a memory such that, after each data carrier has been gripped by the handling device, at least individual or specific data is read from the data carrier held by the handling device and filed in a memory. This allows for a logging of the data carrier associated with the respective registration plate. In this manner it is possible, for example, to create a complete production record of the registration plate. Alternatively, it is also possible to inspect the function of the respective data carrier while at least some of its data are being read.

(15) In an advantageous development, the method can be configured or further modified such that a calculation is made before the handling device supplies data carriers to the receptacles to be provided with a data carrier in or on the plate body or on the material web, preferably by a measuring device or imaging devices. In particular, this involves a separate imaging device. As a result, the handling device can transfer the or each data carrier it holds from the known position exactly to the position or site to be assumed by the data carrier in the respective receptacle of the plate body or material web. Thus, the handling device holding the data carrier can supply the plate

body or material web with the at least one data carrier on the receptacle precisely at the intended position.

(16) A further advantageous development of the method, which can also be an independent solution of the stated object, is characterized in that the data carrier is pre-fixed in the receptacle of the plate body when it is inserted by the handling device into the receptacle of the plate body or the respective receptacle in the material web for the formation of plate bodies, preferably by being placed on at least one adhesion site on or in the respective receptacle such that the data carrier is pre-fixed in the receptacle, which is configured as a depression or opening, for example, when the data carrier is inserted in said receptacle, preferably by being placed on at least one adhesion site previously assigned to the receptacle. The data carrier is then placed or pressed on the at least one adhesion site by the handling device at the precise position relative to the receptacle. Here the data carrier forms an adhesive bond with the adhesion site, which adhesively pre-fixes the data carrier in the exact, non-displaceable position on the at least one adhesion site on or in the plate body or material web. The plate body or the material web and the at least one pre-fixed data carrier can then be subjected to further processing steps without the data carrier changing its position relative to the receptacle during or before such further processing.

(17) Further to an advantageous further possible development of the method, the handling device inserts the data carrier in a respective receptacle, preferably configured as an opening in the plate body or material web, in such a manner that the data carrier, preferably a circumferential edge of it, is spaced from the contours of the opening. Preferably, the data carrier is inserted into the opening such that a circumferential edge of it is evenly spaced apart from the contours of the opening and/or a circumferential gap remains or is formed between the contours of the opening and the edge of the data carrier. This prevents any contact between the data carrier and the plate body or material web formed preferably from a conductive material. In other words, the data carrier is electrically isolated from the conductive plate body or the conductive material web. This is an important prerequisite for a reliable and error-free reading of the data stored on the data carrier, specifically at a relatively great distance.

(18) In another advantageous further possible development of the method, the plate body or the material web is provided, from the rear side, with the at least one adhesion site and/or supplied with the at least one data carrier and/or the respective data carrier is cast or recast in its receptacle. Said measures are thus carried out from the rear side of the plate body or material web. From this position the opening is accessible and covered by the film on the front side of the plate body visible to the viewer after the license plate has been mounted. The plate body or the material web are therefore provided with the at least one data carrier in reverse, so to speak. If however, the receptacle is not formed by a continuous opening but rather by a depression originating at the front side of the license plate, the license plate can be provided with the respective data carrier the right way round, that is to say, with the front side facing upwards. In that case, the film, inasmuch as one is necessary or provided, must be subsequently applied to the front side of the license plate if namely the depression is to be supplied with the data carrier and recast if necessary.

(19) A device for achieving the object stated at the outset is a device for providing a registration plate, preferably a motor vehicle license plate, with at least one data carrier which can be read contactlessly, with at least one work station and with at least one plate body having at least one receptacle for a data carrier or with a material web for the formation of plate bodies of the registration plate, characterized in that a plurality of work stations arranged in succession is provided and at least one conveyor for interlinking the work stations is provided. This device has a plurality of preferably successive work stations and one or also a plurality of conveyors for interlinking at least one, preferably all work stations. This device makes it possible to conduct preparatory steps for the insertion of a data carrier into a receptacle of the respective plate body or the material web, to supply the plate body or the material web with the respective data carrier while pre-fixing the latter during this process, and to fix the data carrier in the receptacle permanently. All

of these steps can be executed with the device in successive cycles.

(20) The relative position of the respective plate body or of the material web relative to the conveyor is reliably maintained during the further transport from one work station to the other, and in particular in the respective work station itself. As a result, the device is particularly suited for the automatic fitting of plate bodies or material webs with data carriers with the assistance of handling devices, measuring devices, positioning aids and/or imaging devices. The automatic fitting of the plate bodies or material webs can therefore be executed rapidly and precisely without any accompanying manual activities.

(21) It is preferably provided that at least one, in particular all, work stations are associated with means, preferably imaging devices such as cameras, in order to identify the respective plate body or material web and/or to assign the respective receptacle to be fitted. As a result, the exact position of each plate body or of the material web, and in particular its receptacle, is known at those work stations where this position is required. This applies in particular to those work stations where a receptacle is fitted with a data carrier. But even at locations where the receptacle is provided with at least one adhesion site and/or the data carrier is permanently fixed by means of casting or moulding, it is important that the exact position of the plate body, material web and/or the respective receptacle on the conveyor is known in order that the handling devices or the like can be employed for the stated purposes.

(22) In an advantageous embodiment of the device a first work station is assigned at least one dispenser device, preferably on a handling device. As an alternative or in addition, it is provided that the first work station is assigned a device, above all an imaging device, for determining the position of the plate body and/or a respective receptacle on the conveyor. In this way, at least one adhesion point can be applied in the receptacle or on the receptacle mechanically at the first work station. This can be realized in a fully automatic manner if the first work station is equipped with a handling device and/or an imaging device or some other means for determining the position of the plate body, the material web and/or the respective receptacle on the conveyor.

(23) Instead of an imaging device at the first work station, the at least one conveyor can be assigned at least one stop at a defined position. In conjunction with at least one sensor capable of detecting the respective stop, the plate body, in particular the material web, can be held ready in a defined relative arrangement on the conveyor, on a predetermined and/or prescribed position of the first work station or in front of or to the side of the latter and/or at further work stations as necessary. The first work station then no longer needs to be assigned any imaging device for determining the position of the plate body, the material web and/or the respective receptacle to be fitted.

(24) A second work station of the device is equipped with stocking means for store of data carriers. This stocking means is preferably realized as a pre-supply conveyor for keeping a number of data carriers ready or is provided with such. The pre-supply conveyor keeps preferably a plurality of in particular flat-lying data carriers ready for processing. Other types of stocking means such as trays or carrier straps or bands are also conceivable.

(25) The second work station preferably has a handling device which is configured to grip in each case a single data carrier or a plurality of data carriers at the same time or in succession from a stocking means, in particular from the pre-supply conveyor. At the same time, the handling device is employed to fit the receptacles with one data carrier each. However, the invention is not restricted to the application of registration plates with only one data carrier. The invention is also suited for fitting each registration plate with a plurality of data carriers if required, preferably simultaneously, but also in succession if necessary.

(26) The second work station is preferably equipped with a stocking means having preferably one pre-supply conveyor and at least one handling device. This increases the degree of automation of the second work station.

(27) The second work station can be further developed by including at least one imaging device,

for example at least one camera. The imaging device provides the handling device with the position of the respective data carrier to be transferred from the stocking means or from the pre-supply conveyor. As a result, the handling means can travel directly to individual data carriers in stocking means or on the pre-supply conveyor in a targeted manner and pick or grip them there. This preferably involves a data carrier among a plurality of data carriers in or on the stocking means, preferably on the pre-supply conveyor, which can be best gripped and/or which can be picked in the easiest or quickest manner.

(28) In particular the second work station has yet a second imaging device which determines the exact position of the data carrier on the handling device after it has been gripped by the handling device, in particular representing this position and/or calculating the position data of the data carrier or providing such a calculation. This enhances the fitting of the plate body or of the material with a respective data carrier in the exact position, in particular the positioning or placing of the data carrier in its exact position on or in the at least one receptacle or in the at least one recess, configured as an opening, in the plate body or material web.

(29) In an advantageous embodiment of the second work station, the handling device associated with it has at least one suction device for pneumatically retaining the or each gripped data carrier by means of negative pressure. The suction device holds in each case a single data carrier in a careful and reliable manner in that it exerts no mechanical forces on the data carrier which might have an adverse effect on it.

(30) If necessary, the second work station can be assigned a third imaging device which establishes the exact position of the plate body to be fitted with the data carrier on the conveyor bearing or holding ready this plate body and in particular provides data for a precise calculation of the position of the relevant plate body and/or its receptacle for the data carrier and makes this data available to the handling device. Such a third imaging device can be dispensed with if receptacles are to be fitted with data carriers in a preferably continuous material web.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Preferred exemplary embodiments of the invention will be discussed in more detail in the following on the basis of the drawing, in which

(2) FIG. 1 shows a perspective representation of a motor vehicle license plate (not yet covered with film),

(3) FIG. 2 shows a partial cross-section through the motor vehicle license plate in FIG. 1, on an enlarged scale,

(4) FIG. 3 shows a perspective total view of the device according to the invention,

(5) FIG. 4 shows a side view of a first work station of the device in FIG. 3,

(6) FIG. 5 shows a top view of the work station in FIG. 4,

(7) FIG. 6 shows a side view of a second work station of the device in FIG. 3, and

(8) FIG. 7 shows a top view of the work station in FIG. 6.

### DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(9) The device shown as a whole in FIG. 3 has three work stations **10**, **11** and **12**, which are each assigned different individual operations. In the shown exemplary embodiment, the three work stations **10**, **11** and **12** are interlinked by two parallel, preferably identical conveyors **13**, which are realised as belt conveyors. However, the invention can also be realized with different conveyors, conveyors running in different courses, more than two conveyors and/or with only a single conveyor **13**.

(10) The device is employed to provide, preferably to fit, registration plates, in particular motor vehicle license plates, with data carriers **14** that can be read and/or written contactlessly. This may



involve RFID chips which operate at various frequency ranges, but also NFC (Near Field Communication) chips, for example. In the present exemplary embodiment, it will be assumed that each registration plate is to be fitted with a single data carrier **14**. However, the invention is also suited for fitting identification tags with a plurality of identical or even different data carriers **14**.

(11) In the case of the exemplary embodiment shown here, the data carrier **14** is provided, in particularly connected, to a plate body **15** of the license plate made of an electrically conductive material, such as aluminium or, alternatively, a synthetic material that has been made at least partially electrically conductive. For this purpose, the plate body **15** has a receptacle for the data carrier **14**. In the present exemplary embodiment, the receptacle is formed as a punched-out opening **16** in the plate body **15**, for example. However, the receptacle can also be a depression in the plate body **15**. The opening **16** extends uninterrupted with the same cross-section from a front side **37** to the rear side **39** of the plate body **15** (FIG. 2). The opening **16** has a base area corresponding to the contour of the data carrier **14**. However the opening **16** is somewhat larger than the base area of the data carrier **14**, so that the outer edge **17** of the data carrier **14** is spaced all around, preferably evenly, from the contour **18** of the opening **16** after the data carrier **14** has been inserted in the opening **16**. This produces a gap **19** running all the way round between the data carrier **14** and the opening **16**. The ring-shaped gap **19** is preferably of uniform width at all points.

(12) In the shown license plate the plate body **15** has an elongated slit **38**. The slit **38** leads to a narrow side in the opening **16**, thus forming a transition from the slit **38** to the opening **16**. The slit **38** forms in particular at least one part of an antenna for the data carrier **14**.

(13) The data carrier **14** is permanently connected to the plate body **15** in the opening **16** by means of casting with a castable, non-conductive casting compound, such as a castable plastic and/or synthetic resin which preferably hardens or cures with UV radiation. Here the data carrier **14** can be completely embedded in the casting compound so that the casting compound not only fills out the circumferential gap **19** but also covers the complete surface of the front side **37** and/or rear side **39** of the data carrier **14**. If necessary, one side of the data carrier **14** can remain uncovered by the casting compound or casting material.

(14) In the shown exemplary embodiment the plate body **15**, specifically the front side **37** (the side facing up in FIG. 2) of the plate body **15** visible in its assembled state, is covered or laminated with a thin film **20**, for example a self-adhesive film. The film **20** extends at least across the surface area of the opening **16** and an adjacent circumferential region of same. If required, the film **20** covers the entire bottom side of the plate body **15** (FIG. 2).

(15) The first work station **10** is configured for the purpose of attaching a single adhesion site **21**, but also a number of adhesion sites **21** if necessary. The adhesion site **21** can be a small-area casting compound or adhesive point, for example. The adhesion site **21** is applied in the contours of the opening **16** or, alternatively, in a trough-like depression, preferably in the centre of the area. The application of the adhesion site **21** in the opening **16** or depression results in the adhesion site **21** being positioned on the film **20** freely exposed in the region of the opening **16**. The adhesion site **21** is then located on the film **20** inside of the opening **16**. The adhesion point **21** can be formed from a hot glue or a casting compound capable of solidifying, for example, which still exhibits adhesive characteristics after application for the purpose of pre-fixing the data carrier **14**.

(16) The first work station **10** can be dispensed with if the adhesion site **21** is formed by the adhesive layer on the bottom side of the film **20** (FIG. 2) facing the interior of the opening **16** because then no separate adhesion site **21** needs to be applied.

(17) Following the first work station **10**—inasmuch as it is not dispensable—in the direction of transport **22** of the conveyor **16** is the second work station **11**. In this work station **11** the plate body **15** is fitted with the data carrier **14** by the automatic insertion of the data carrier **14** in the opening **16**. In the process the bottom side of the data carrier **14** is placed on the adhesion site **21** in the opening **16** or directly on the film **20** and pressed upon it if necessary in order to pre-fix the data carrier **14** in the opening **16** of the plate body **15**.

(18) The third work station **12**, which follows the second work station **11** in the direction of transport **22**, serves to embed and/or recast the data carrier **14** in the opening **16** of the plate body **15**, thus protectively accommodating the data carrier **14** in the opening **16** and fixing it permanently in the opening **16** to achieve a non-detachable and damage-free connection of the data carrier **14** to the plate body **15**. In addition, the casting or embedding of the data carrier **14** in the opening **16** results in insulating the data carrier **14** from the electrically conductive plate body **15**.

(19) Located upstream of the first work station **10** is a magazine (not shown) with a supply of a plurality of plate bodies **15**. The magazine can be realised such that it can accommodate batches of plate bodies having different formats. Individual plate bodies **15** are automatically taken from the magazine one after the other, for example by means of an appropriate charging device, and placed on upper strands **23** of the conveyor **13**, preferably in an alternating manner. Depending on the configuration of the device, this can occur at specific positions on the upper strand **23** as established by stops and/or carriers for the plate bodies, for example, but also in an approximate position, as an alternative. However, the orientation of the plate body **15**, particularly with respect to its opening **16**, should preferably always be the same.

(20) The plate bodies **15** are inversely positioned on the upper strand **23** of the conveyor **13**, in other words with the rear side **39** facing upwards, in order that the openings **16** in the plate bodies, which are covered by the film **20** on the front side **37** of the plate body **15**, are freely accessible from the rear side **39** of the plate body **15**. By virtue of the openings **16** in the plate bodies **15** being freely accessible at the rear side, adhesion sites **21** can be introduced into the openings **16** and the openings **16** can be fitted with data carriers **14**. In addition, the openings **16** open at the rear can be casted as such.

(21) The first work station **10** is equipped with a handling device **24**, which bears a dispenser device **25** at the end of its arm. The dispenser device **25** is realised such that it can release in a targeted manner a drop of a flowable casting material for the formation of the adhesion site **21**. The dispenser device **25** is assigned a stocking means and, in addition, a heating device in case the glue or adhesive being used is a type of hot glue.

(22) The first work station **10** is in addition equipped with an imaging device **26**, for example a camera, which may be a two- or three-dimensional camera, which can be either a black-and-white camera or a colour camera. The imaging device **26** determines the position of the opening **16** of the plate body **15** held ready on the upper strand **23** of the respective conveyor **13** in the first work station **10**. Image data recorded by the imaging device **26** are transferred to a control system of the handling device **24**, where the position of the opening **16** relative to the handling device **24** and/or to the dispenser device **25** is calculated from the data. On the basis of the position thus calculated, the handling device **14** can move the dispenser device **25** into the position designated for the application of the adhesion site **21** at the opening **16** and apply the adhesion site **21** on the film **20** preferably at the centre of the opening **16**.

(23) The second work station **11** also has a central handling device **27**. Located at the free end of a gripper arm **28** of the handling device **27** is at least one suction device **29** charged with negative pressure **29** and with an elastically deformable suction pad. The suction device **29** is configured and dimensioned such that it holds with negative pressure a flat side, specifically the top side, of a data carrier **14**.

(24) In the shown second work station **11**, the handling device **27** is arranged somewhat between the inner conveyor of the two parallel conveyors **13** running adjacent to one another and a pre-supply conveyor **30** of a stocking device **31**. The stocking device **31** contains a stock of a number of preferably identical data carriers **14**. Data carriers **14** are taken one by one from this stock to an upper strand **32** of the pre-supply conveyor **30** which is realised as a belt conveyor. There is always a plurality of data carriers **14** lying on the upper strand **32** with arbitrary orientation. Here the top or bottom sides of the data carriers **14** can lie facing upwards or downwards at random. The conveyor belt of the pre-supply conveyor **30** can be driven preferably continuously at a low speed such that

the upper strand **32** of the conveyor **13** continues to move along with the data carriers **14** upon it contrary to the direction of transport **22**.

(25) The second work station **11** has a first imaging device **33** in the region of pre-supply conveyor **30** above its upper strand **32**. Said imaging device records an image of all data carriers **14** on the upper strand **32** or at least those located in the action area of the gripper arm **28** of the handling device **27**. The imaging device **33** also records which side of the data carrier **14** lies upwards. Following the evaluation of the data transferred from the imaging device **33** to the control system of the stocking device **31**, at least one suction device **29** of the handling device **27** is driven, as determined by the data of the imaging device **30**, to the position of such a data carrier **14** which can be easily gripped and which is suited for fitting the plate body **15** because it lies the right way round on the upper strand **32** of the pre-supply conveyor **30**, that is to say, with its top side actually facing upwards, and which is also intended to lie facing upwards after the plate body **15** has been fitted.

(26) Data carriers **14** which lie the wrong way round on the upper strand **32** of the pre-supply conveyor **30** are not gripped and transported by the pre-supply conveyor **30** back to the start of the pre-supply conveyor **30** or stocking device **31**. They continue to circulate and, if oriented correctly, return to the start of the pre-supply conveyor **30**.

(27) The drive of the pre-supply conveyor **30** can be temporarily shut down for the visual recording of the data carriers **14** and for the suction action to be applied to a selected data carrier **14** by the suction device **29** of the handling device **24**. Provision is preferably made that the pre-supply conveyor **30** is driven continuously, specifically at a constant uniform speed which the control system of the handling device **27** can take into account or compensate.

(28) The handling device **27** is preferably assigned a read station **34** near the pre-supply conveyor **30**. After a data carrier **14** has been gripped by the suction device **29** of the handling device **27**, this read station **34** serves to read in a contactless manner at least one item of information already stored on the data carrier **14**. For example, this can be a serial number of the data carrier **14**, but also, as an alternative or in addition, at least one other specification, such address data. On the basis of this information, the license plate can be identified once the plate body **15** has been fitted with the data carrier **14**.

(29) Furthermore, the second work station **11** can have a testing device which checks the data carrier **14** gripped by the handling device **27** as to whether it is in order, in particular if it can be read.

(30) The testing station is preferably integrated into the read station **34** so that the reading of at least one item of information from the gripped data carrier **14** is simultaneously linked with its testing. For example, the non-readability of the desired information indicates a faulty function of the data carrier **14**. Such a data carrier **14** is unusable and is removed as a reject by being released by the handling device **27**.

(31) Following or, if necessary, even during the readout of the desired data from the carrier **14**, the data carrier **14** retained by the suction device **29** of the handling device **27** is recorded by a second imaging device **35** with respect to its exact position under the suction device **29**, in other words on the handling device **27**. This imaging device **35** is arranged between the pre-supply conveyor **30** and the conveyor **13** opposite thereto, specifically below the suction device **29**. In this way, the data carrier **14** held by the suction device **29** of the handling device **27** passes by the imaging device **35** on its way to the conveyor **13**. As it moves past the imaging device **25**, or briefly pauses at the latter, the imaging device **35** measures the relative position of the data carrier **14** held by the suction device **29**. As a result, it is possible to establish, for example, the position at which the data carrier **14** is held by the handling device **27** as expressed in a coordinate system running through the vertical longitudinal centre axis of the suction device **35**. The data of the second imaging device **35** transferred to a control system of the handling device **27**, specifically the acquired position values, provide the handling device **27** with the first reference values for the insertion of the data

carrier **14** into the opening **16** of the plate body **15** in the precise position.

(32) The second work station **11** has at least one further imaging device **36**. In the shown device having two conveyors **13**, each conveyor **13** is assigned its own imaging device **36**. Each of the two imaging devices **36** is disposed above one of the two parallel conveyors **13**. The respective imaging device **36** captures the position of the opening **16** of the plate body **15** lying turned over on the respective conveyor **13**. The plate body **15** that is first brought into a position that is readily transferable for a data carrier **14** has priority in being fitted with the data carrier **14** momentarily held by the handling device **27** ("first in-first out" principle). If, for example, a plate body **15** is on an outer conveyor **13**, the opening **16** is recorded by the imaging device **36** associated with said conveyor. The image data are then transferred to the control system of the handling device **27**, which calculates the exact position of the opening **16** of the plate body **15**. This calculated position of the opening **17** provides the handling device **27** with the second position required for the fitting operation. The data carrier **14** held by the handling device **27** is then positioned by the latter exactly over the opening **16** of the plate body **15** on the outer conveyor **13** and inserted into the opening **16**. This is done by having the data carrier **14** placed, and if necessary pressed down, in the exact position provided in the opening **26** earlier at the first work station **10**, namely onto the adhesion site **21** on the film **20** lying directly on the conveyor **13** under the plate body **15** lying upside down on the conveyor **13**, whereby the adhesion site **21** pre-fixes the data carrier **14** in the position it has been inserted to by the handling device.

(33) When the plate body **15** is fitted with the data carrier **14** in the second work station **11**, the pre-fixing of the data carrier **14** on the plate body **15** is made in particular by the at least one adhesion site **21**. In addition, the automatic fitting process is executed in such a targeted manner that results in the circumferential gap **19**, preferably of equal width, being formed between the edge of the data carrier **14** and the contours **18** of the opening **16**, thus ensuring that the data carrier **14** has no contact with the conductive plate body **15**.

(34) The third work station **12** following the second work station **11** serves to fix the data carrier **14** permanently in the opening **16** of the plate body **15**. For this purpose, in the third work station **12**, the space of the opening **16** not filled in by the plate body **15** is recast with a flowable insulating material, such as a plastic and/or synthetic resin compound which hardens or cures with UV radiation. The space in the opening **16** left empty by the data carrier **14** is thereby filled out, preferably completely. In the process, at least the circumferential gap **19** between the edge **17** of the data carrier **14** and the contours **18** of the opening **16** is filled out, thus producing an all-round bridge-like connection of the data carrier **14** with the plate body **15**. At the same time, the electrically non-conductive casting material serves to form in the gap **19** an insulation barrier all around the data carrier **14** so that the latter has no electrically conductive contact with the plate body **15**. In particular, the casting material also fills out the intermediate space between the film **20** and the bottom side of the data carrier **14** formed by the adhesion site **21** attached to the film **20** in the first work station **10** for the pre-fixing operation in the region of the opening **16**. As a result, the entire bottom side of the data carrier **14** is also wetted and covered with casting material. When the data carrier **14** is cast in the opening **16**, even the top side of the data carrier **14** can be completely covered by the casting compound over its entire surface. The data carrier **14**, enclosed on all sides by the casting compound, is thereby fixed in the opening. Furthermore, the data carrier **14** is completely embedded in the electrically non-conductive casting compound. This also causes the casting compound to be flush with the rear side **39** of the plate body **15**.

(35) The casting of the data carrier **14** in the third work station **12** is carried out by a handling device assigned to the latter. To this end, the handling device, exactly like the handling device **24** of the first work station **10**, is equipped with a dispenser device, which can be supplied with a stored quantity of a flowable and/or castable compound or some other electrically non-conductive substance.

(36) The third work station **12** also has an imaging device. This imaging device records image data

used to calculate the position of the plate body **15** on the respective conveyor **13** and/or of the opening **16** and/or of the data carrier **15** inserted therein. The handling device then moves the dispenser device arranged on it for a flowable casting compound, or a casting compound made flowable by heating, to the position above the data carrier **14** in the opening **16** of the plate body **15** as determined by the imaging device, where a precisely metered quantity of casting compound for casting or recasting the data carrier **14** in the opening **16** is delivered by the dispenser device. The casting compound then flows into the cavity between the opening **16** and the data carrier **14** and fills it out completely before it hardens and/or cures, e.g. under the influence of light.

(37) Finally, the cast plate body **15** fitted with the data carrier **14** is removed by the relevant conveyor **13** to a stacking and/or packaging device for plate bodies **15** provided with data carriers **14**. If necessary, the conveyors **13** can be lengthened downstream of the third work station **12** to form a finishing line. This allows the casting compound to harden sufficiently and cool off if necessary, thus ensuring that the final fixation of the data carrier **14** in the opening **16** of the plate body **15** can no longer become detached.

(38) In place of all or only some of the imaging devices **26**, **33**, **35** and **36** it is possible to employ measuring devices with a different configuration and which do not necessarily involve cameras.

(39) The invention is also not restricted to three interlinked work stations. The number of work stations may be greater or less. For example, the first work station can be omitted if the film **20** is self-adhesive on the side facing the opening **16** because then the application of at least one adhesion site **21** on the film **20** in the region of the opening **16** is no longer necessary. It is also conceivable, for example, to consolidate the second work station **11** and the third work station **12** and/or to interconnect a plurality of work stations.

(40) It is also possible to vary the number of conveyors **13**. It is possible to have more than two conveyors **13** or even just a single conveyor **13**. Likewise, the conveyors **13** do not necessarily have to be realised as belt conveyors. They may also be pocket conveyors, operable roller conveyors or the like.

(41) A second exemplary embodiment of the invention not shown in the drawings provides for fitting a continuous and/or band-shaped material web made of the material of the plate body, in particular aluminium or another such electrically conductive material, with data carriers and permanently connect them with the material webs. Following in succession in the longitudinal direction of the material web, preferably in tight formation, the plate bodies are chained by the material web to each form a license plate. The material web have a width which corresponds to the width of the license plate or which is somewhat wider. The material web can be unwound endlessly and/or continuously from a supply reel.

(42) The material web has recesses, either openings or depressions, at specifically defined positions for one data carrier each. For this purpose, the material web has preferably a single receptacle, if necessary a plurality of receptacles, for each plate body or license plate to be punched from it.

(43) After the receptacles of the material web have been provided with data carriers one by one and attached thereto, blanks, in particular plates, for the plate body of each license plate are punched from the material web in succession. At the same time, or subsequently, the plate body can be provided with a circumferential embossed edge. Later, the intended characters, in particular letters and/or digits as well as other marks as necessary, are embossed into the plate bodies that have been fitted with at least one data carrier.

(44) In the following it will be assumed that registration plates, preferably motor vehicle license plates, each having only a single data carrier, are to be produced from the material web, without this restricting the scope of the invention.

(45) The material web having a receptacle for the data carrier of each license plate is moved along three individual and successive work stations, preferably three work stations pursuant to the exemplary embodiment described previously, specifically in a non-continuous manner with brief standstill periods at the individual work stations. For this purpose, the individual work stations can

be interlinked with a common conveyor as in the previously described exemplary embodiment. But it is also conceivable to provide merely one support running past all work stations, for example a roller track with non-driven rollers. The material web is then further transported by it being pulled along on the base, support or roller track. Here a supply of the material web is preferably drawn off a roll or the like at the same time.

(46) Like the pre-fixing conducted at the first work station and the final fixation, in particular the casting of the receptacles, namely the space left open in the region of the receptacles at the third work station, the fitting of the receptacles of the material web with data carriers at the second work station is in principle executed in the device of this exemplary embodiment exactly as explained in the case of the previously described exemplary embodiment, with references being made to the preceding description of the device and method of the first exemplary embodiment shown in the figures.

(47) In a departure from the previously described exemplary embodiment of the invention, the openings for the data carrier are applied in the material web or material strips at precisely the positions indicated for the subsequent punching out of the license plate or plate body. As it passes the individual work stations the material web or the material strip is guided, for example by means of stops, laterally or in other words transversely to the continuation or transport direction. As a result, the openings are precisely aligned at the individual work stations transversely to the longitudinal direction of the strip-like material web. Due to the exactly positioned openings in the longitudinal direction of the material web, the further transport of same from one work station to the other can be stopped at every work station at the exact position. This can be achieved by measuring devices, for example light barriers as well as signal receivers for driving the further movement of the strip-shaped material web. As a result, the openings in the web or in the material strips of preferably all work stations are held ready in an exact position. For applying the data carriers to the openings, the handling devices can then be brought in a targeted manner to the exactly determined positions of the data carriers in the material web.

(48) Provision is preferably made for spacing the individual work stations in succession from one another so that when the further transport of the material web is stopped in a position determined by a measuring device, for example, the activities assigned to all work stations, specifically at least the application and simultaneous pre-fixing of the respective data carrier in or on the receptacle as well as the final fixation of the data carrier by means of casting or recasting, can be carried out.

(49) When the data carriers are applied in or on the receptacles of the continuous material web, it is not absolutely necessarily for imaging devices located at least at some working stations to record the position of the receptacle where the receptacles are to be provided with data carriers and/or the data carriers are to be cast in the receptacles. Since the receptacles have been provided in the material web at precisely predetermined positions, The measuring devices described previously are also sufficient for positioning the material web at the work stations, preferably at all work stations simultaneously, in order to position the data carriers on or in the receptacles correctly and precisely and for the final and permanent fixation of the data carriers in or on the receptacles with casting compound. In this case, it is not absolutely necessary to provide handling devices at least in the first and/or third work station.

(50) In case the bottom side of the material web is already coated with a self-adhesive film and the receptacle is realised as a continuous opening in the material web, it is possible to dispense with, if appropriate, the application of at least one adhesion site at the first work station as described previously with respect to the first exemplary embodiment. Then the first work station does not have to be present in this second exemplary embodiment.

(51) If in the case of this exemplary embodiment a first work station for the application of at least one adhesion site of glue or a casting compound is still provided, the at least one adhesion site does not necessarily have to be applied by a handling device. On account of the receptacle in the material web which is always held ready at the same position at the work station, it may be

sufficient to provide at least one dispenser device at this position at the first work station, preferably a stationary dispenser device.

(52) The third work station or—if the first work station is not present—the last work station also does not necessarily need to be equipped with a handling device for casting the data carrier applied in the previous work station because the material web is also halted at this last work station such that the respective receptacle with the already applied data carrier is always located exactly at the same position. There is then no need to provide any handling device at this last work station. It can suffice to provide merely at least one dispenser device for the casting compound, preferably a stationary one.

#### LIST OF DESIGNATIONS

(53) **10** first work station **11** second work station **12** third work station **13** conveyor **14** data carrier **15** plate body **16** opening **17** edge **18** contour **19** gap **20** film **21** adhesion site **22** direction of transport **23** upper strand **24** handling device **25** dispenser device **26** imaging device **27** handling device **28** gripper arm **29** suction device **30** pre-supply conveyor **31** stocking device **32** upper strand **33** imaging device **34** read station **35** imaging device **36** imaging device **37** front side **38** slit **39** rear side

## Claims

1. A device for providing a registration plate with at least one data carrier which can be read contactlessly, the registration plate being made from a plate body having at least one receptacle for the at least one data carrier or from a material web for the formation of plate bodies, wherein the device comprises a plurality of work stations arranged in succession and at least one conveyor for interlinking the work stations, wherein at least one of the work stations is assigned a means for determining the position of the plate body or of the material web for the formation of plate bodies and/or of the at least one receptacle on the at least one conveyor, wherein a first of the work stations and/or a third of the work stations is assigned a handling device having at least one dispenser device, wherein the first of the work stations and/or the third of the work stations is assigned the means for determining the position of the plate body and/or of the at least one receptacle or of the material web for the formation of plate bodies on the conveyor, wherein a second of the work stations is assigned a stocking device for the provisioning of a plurality of data carriers, and the second of the work stations has a handling device which is configured for the purpose of gripping at least one single one of the plurality of data carriers from the stocking device, and the second of the work stations is configured for placing the one of the plurality of data carriers into a predetermined receptacle of the plate body or material web, wherein the second of the work stations has a measuring or imaging device which supplies the handling device of the second of the work stations with the position of the data carrier to be gripped at the stocking device, or the second of the work stations has a measuring or imaging device for determining the position of the data carrier after the data carrier has been gripped by the handling device of the second of the work stations, wherein the handling device of the second of the work stations has at least one suction device for pneumatically retaining the or each of the gripped data carriers by means of negative pressure, wherein the registration plate is a motor vehicle license plate.

2. The device as claimed in claim 1, wherein the stocking device has a pre-supply conveyor or is configured as a pre-supply conveyor.

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